Many Daughters – Pre Analysis Plan

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Introductory Remarks

Via the instructions of the project our choice of data is limited to the German Socioeconomic Panel and the yearly waves between 2000 and 2021. The guiding principle for our subsequent modeling decisions is our expectation of a strong selectivity into parenthood which we cannot control for. Further, we assume that, in contrast to the general event of childbirth, having a daughter versus a son is a quasi-random event resembling a natural experiment. By using the parents of daughters as a treatment and parents of sons as a control group, we isolate the effect of having a daughter from having a child in general. This prerequisite enables us to mostly refrain from the use of control variables, since it implies control and treatment group are similar in their composition. The one control variable we include is maternal age at birth, because it has been shown to be associated with the sex of the child[1]. We use sampling weights in the case of selective panel attrition based on the child's gender. We focus on first births, because selectivity into higher order births may already be systematic, e.g. having children until a boy is born. We restrict on children born in the two years preceding the wave of measurement of the dependent variable to have children of similar age. We control for further children born in that observation window. The randomness of child gender enables us to apply OLS-regression analyses to dependent variables measured at one time point. From a pragmatic point of view, this is an advantage, since some of the concepts (e.g. political preferences) are seldomly measured in the SOEP data and would not allow for panel regression methods.

General Inclusion and Exclusion Criteria

We limit our sample to individuals who became first time parents in the yearly waves between 2000 and 2021. Our sample includes only German citizens. Cases with missing values on the respective dependent variables, as well as the selection variables (daughter or son) and control variables are listwise deleted. The exact samples vary along the hypotheses, as we focus on parents who had their first child in one of the two waves preceding the measurement of the dependent variable.

Variables included in all models

- BirthYearFirstChild: kidgeb01
    - Coding: Negatives to missing
- BirthYearFirstChild: kidsex01
    - Coding (original): (1) Male (2) Female, missing values are negative
    - Coding (new): Negative to missing, Male (0) Female (1)
- SexSecondChild
    - This variable indicates whether a second child was born before the outcome was measured.
    - We use four variables to construct this variable:
        - Year Of Birth 2nd. Child: kidgeb02
            - Original coding: Numeric, missings -1, -3, -4, -5, -6, -7, not applicable -2
        - Sex 2nd. Child: kidsex02:

[1] Matsuo, K., Ushioda, N., & Udoff, L. C. (2009). Parental aging synergistically decreases offspring sex ratio. *The journal of obstetrics and gynaecology research*, *35*(1), 164–168. https://doi.org/10.1111/j.1447-0756.2008.00836.x
Rueness, J., Vatten, L., & Eskild, A. (2012). The human sex ratio: effects of maternal age. *Human reproduction (Oxford, England)*, *27*(1), 283–287. https://doi.org/10.1093/humrep/der347

        - Original coding: (1) Male (2) Female, missing values are negative
    - Month Of Birth 2nd. Child: kidmon02
        - Original coding: (1) to (12), missing values are negative
    - Month Of Interview: imonth
        - Original coding: (1) to (12), missing values are negative
- New variable: We set 1, -3, -4, -5, -6, -7 on kidgeb02 to missing, and on the other three variables, all negatives to missing. We drop any cases with missings on any of the variables. We then recode those where kidgeb02 is not applicable (-2, no second child) to zero. Using imonth (survey month) and kidmon02 (month of birth of second child), we then determine whether there was a second child up to one month before the measurement of the outcome (2018 or 2019, depending on hypothesis). If there was a second child, we then use kidsex02 to code sons to 1 and daughters to 2. We end with (0) No second child within the time frame, (1) second child son, and (2) second child daughter. The construction of the variable is the same for all hypotheses, but since the time frames vary by one year, the values on this variable may differ between hypotheses.

- MotherAge: gebjahr and kidgeb01
    - We use birth year of the mother (gebjahr) and birth year of the first child (kidgeb01) to construct a variable indicating the age of the mother at birth of the first child.

Hypothesis 1: Having daughters leads to stronger support for gender equality.

Inclusion and Exclusion Criteria

We include those who had their first child in 2016 or 2017 and have no missing information on our outcome measure (bip_197_04), which requires SOEP participation in 2018.

Analytical approach

$$Y_i = \beta_0 + \beta_1 Daughter + \beta_3 BirthYearFirstChild + \beta_4 Daughter * BirthYearFirstChild + \beta_5 MotherAge + \beta_6 SexSecondChild + \epsilon_i$$

- *Y* represents support for gender equality as measured by our outcome variable
- *Daughter* is a binary indicator for the gender of the first child
- *BirthYearFirstChild* is an indicator whether the child was born in 2016 or 2017
- *MotherAge* is a numeric indicator indicating the age of the mother at the time of birth of the first child
- *SexSecondChild* is a categorical variable remaining at zero for those who did not have a second child at the time of the outcome measurement, 1 for those who had a son, and 2 for those who had a daughter.
- We robust standard errors clustered at the household level and use sampling weights provided by the SOEP.
- ➢ In assessing whether we consider the hypothesis confirmed, we use STATA's margins command to predict the outcomes at both levels of *Daughter* while keeping *SexSecondChild* at zero. Other covariates are kept at means. We will also look at the significance for both birth years (2016 and 2017) separately. We consider results statistically significant at the 0.05 level.

Variables

- Children below the age of 6 suffer if their mother works: bip_197_04
    - Coding: (1) Completely Disagree to (7) Fully Agree. Negatives will be recoded to missing and coding will be reversed so that (1) Fully Agree and (7) Completely Disagree for interpretability reasons.
- See above: *Variables included in all models*

Hypothesis 2: Having daughters leads to more left-leaning preferences

Inclusion and Exclusion Criteria

We include those who had their first child in 2017 or 2018 and have no missing information on our outcome measure (bjp_171), which requires SOEP participation in 2019.

Analytical approach

$$Y_i = \beta_0 + \beta_1 Daughter + \beta_3 BirthYearFirstChild + \beta_4 Daughter * BirthYearFirstChild + \beta_5 MotherAge + \beta_6 SexSecondChild + \epsilon_i$$

- *Y* represents political attitudes on a left-right spectrum
- *Daughter* is a binary indicator for the gender of the first child
- *BirthYearFirstChild* is an indicator whether the first child was born in 2017 or 2018
- *MotherAge* is a numeric indicator indicating the age of the mother at the time of birth of the first child
- *SexSecondChild* is a categorical variable remaining at zero for those who did not have a second child at the time of the outcome measurement, 1 for those who had a son, and 2 for those who had a daughter.
- We robust standard errors clustered at the household level and use sampling weights provided by the SOEP.
- ➢ In assessing whether we consider the hypothesis confirmed, we use STATA's margins command to predict the outcomes at both levels of *Daughter* while keeping *SexSecondChild* at zero. Other covariates are kept at means. We will also look at the significance for both birth years (2017 and 2018) separately. We consider results statistically significant at the 0.05 level.

Variables

- Political Attitude: bjp_171 (2019)
- **Coding:** 0 (completely left) to 10 (completely right). Negatives will be recoded to missing and coding will be reversed so that (0) completely right and (10) completely left for interpretability reasons.
- See above: *Variables included in all models*

Hypothesis 3: Having daughters leads to more gender equality in the household (as measured by behavior or outcomes)

Inclusion and Exclusion Criteria

In ensuring comparability with other hypotheses, we include individuals who had a first child in 2017 or 2018. However, in this case, we restrict our analysis to heterosexual couples who remain together for the duration of the observation period, since our outcome measure is the gender care gap within the couple. For this hypothesis, households, not individuals, are the unit of analysis.

Analytical approach

$$Y_c = \beta_0 + \beta_1 Daughter + \beta_3 BirthYearFirstChild + \beta_4 Daughter * BirthYearFirstChild + \beta_5 MotherAge + \beta_6 SexSecondChild + \epsilon_c$$

- *Y* represents the gender care gap within the couple in 2019 (Gap_c)
- *Daughter* is a binary indicator for the gender of the first child
- *BirthYearFirstChild* is an indicator whether the first child was born in 2017 or 2018
- *MotherAge* is a numeric indicator indicating the age of the mother at the time of birth of the first child
- *SexSecondChild* is a categorical variable remaining at zero for those who did not have a second child at the time of the outcome measurement, 1 for those who had a son, and 2 for those who had a daughter.
- We use sampling weights provided by the SOEP.

➢ In assessing whether we consider the hypothesis confirmed, we use STATA's margins command to predict the outcomes at both levels of Daughter while keeping SexSecondChild at zero. Other covariates are kept at means. We assess whether the differences in predicted outcomes are statistically significant at the 0.05 level. We will also look at the significance for both birth years (2017 and 2018) separately.

Variables

- Within-Couple Gender Care Gap
  - bjp_05_04 (2019): Number of hours spent looking after children on an average weekday
  - bjp_05_03 (2019): Number of hours spent on housework (e.g., washing, cooking, cleaning) on an average weekday
  - Negative values are recoded as missing.
  - Couples are identified using the SOEP partner identifier (*partnr*), which links the responses of both partners.
  - We calculate the couple-specific gender care gap as:

$$Gap_c = \frac{(\text{bjp_05_04}_m + \text{bjp_05_03}_m) - (\text{bjp_05_04}_w + \text{bjp_05_03}_w)}{(\text{bjp_05_04}_m + \text{bjp_05_03}_m)} \; x\; 100$$

    - Suffix *m* denotes the hours devoted by the male partner
    - Suffix *w* denotes the hours devoted by the female partner
    - Gap_c expresses the percentage difference in care work within the couple, relative to the man's total hours.
  - See above: *Variables included in all models*

Hypothesis 4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes)

Inclusion and Exclusion Criteria

We include those who had their first child in 2016 or 2017 and have no missing information on our outcome measure (bip_197_04), which requires SOEP participation in 2018.

Analytical approach

For this hypothesis, we run models that are analogous to hypothesis (1), but separate by gender.

Model for women:

$$Y_i^W = \beta_0 + \beta_1 Daughter + \beta_3 BirthYearFirstChild + \beta_4 Daughter * BirthYearFirstChild + \beta_5 MotherAge + \beta_6 SexSecondChild + \epsilon_i$$

Model for men:

$$Y_i^M = \beta_0 + \beta_1 Daughter + \beta_3 BirthYearFirstChild + \beta_4 Daughter * BirthYearFirstChild + \beta_5 MotherAge + \beta_6 SexSecondChild + \epsilon_i$$

- *Y* represents support for gender equality as measured by our outcome variable
- *Daughter* is a binary indicator for the gender of the first child
- *BirthYearFirstChild* is an indicator whether the first child was born in 2016 or 2017
- *MotherAge* is a numeric indicator indicating the age of the mother at the time of birth of the first child
- *SexSecondChild* is a categorical variable remaining at zero for those who did not have a second child at the time of the outcome measurement, 1 for those who had a son, and 2 for those who had a daughter.
- We use sampling weights provided by the SOEP.
- ➢ In assessing whether we consider the hypothesis confirmed, we use STATA's margins command to predict the outcomes at both levels of Daughter while keeping SexSecondChild at zero. Other covariates are kept at means. We will also look at the significance for both birth years (2016 and 2017) separately. We consider results statistically significant at the 0.05 level.

Variables

- Children below the age of 6 suffer if their mother works: bip_197_04
    - Coding: (1) Completely Disagree to (7) Fully Agree. Negatives will be recoded to missing and coding will be reversed so that (1) Fully Agree and (7) Completely Disagree for interpretability reasons.
- See above: *Variables included in all models*